

Hardware-in-the-loop (HIL) Integrated Design Platform for High-frequency Controller Development of WBG Power Converters

Ravi Nath Tripathi, Hirotugu Matoba

Nagamori Actuator Research Center, Kyoto University of Advanced Science, Japan

Email: tripathi.ravi@kuas.ac.jp

Abstract —Wide bandgap (WBG) power devices have challenges such as fast and accurate device models and high-frequency feedback control considering design and prototyping of power converter systems. In this paper, an integrated hybrid environment is used for GaN-based non-inverting buck-boost (NIBB) converter design with high-frequency feedback control. The hybrid environment consists of PLECS, and Xilinx Vitis Model Composer integrated into MATLAB. The circuit model of the GaN-based converter developed in PLECS, and the high-frequency feedback control modelled in Xilinx Vitis Model Composer. The entire system model was validated through the FPGA Hardware-in-the-loop (HIL) simulation. Further, FPGA-based feedback control validated through experimental tests for GaN dc-dc converter system.

I. INTRODUCTION

The power electronic system is one of the most vital parts for the electrical instruments. The electrification and advancement in technologies makes it the core of home appliances, electrical vehicles, and renewable energy systems. The advent of wide-bandgap (WBG) devices leads with more fascination of power electronic systems considering downsizing of the overall system, better efficiency, and power density because of the operational capability at higher switching frequency. Due to this, WBG GaN power devices becoming one of the major choices for different power applications such as data centers, computers/laptops, chargers, and LEDs.

The GaN devices possessing the higher switching speed as compared to the Si and SiC devices [1]. And the higher switching speed ultimately provides the operational capability at higher switching frequency comparatively with very low switching losses. High switching frequency operational capability lower the ripples in the output voltage/current and exploited to have the size reduction in passive components: inductors and capacitors. However, power converter operation at high switching frequency is crucial considering the feedback control loop. The generation of appropriate switching signals for the desired operation of the power converter system achieved through the feedback control loop.

The implementation of a feedback control loop is crucial considering the possible switching of the WBG devices at the higher switching frequency and control bandwidth. The fast feedback control loop in conjunction with a high controller bandwidth system is achievable through the FPGA-based system development [2]. Consequently, this updates the duty after every switching cycle and provides appropriate switching signals that result in improved controller performance.

FPGA-based feedback control system development involved the intricacy of HDL code. This involves the complexity of control algorithms to implement feedback control concerning the high switching frequency operation of WBG devices. The system modeling approach with

integrated model-based design (MBD) used to automatically generate the HDL code for FPGA controller boards.

The dc-dc converters have a key part in power applications to perform the step-up/step-down of the dc voltage according to the system requirement. Non-inverting buck-boost (NIBB) converter fulfils the requirement of both: step-up and step-down of dc voltage without change in the output voltage polarity. This paper presents a hybrid integrated platform for feedback control development of WBG converters. In this paper, a high-frequency feedback control loop is designed and developed for the GaN-based non-inverting buck-boost (NIBB) dc-dc converter. The thermal model of the GaN device is used to design the NIBB converter using the PLECS-blockset through the MATLAB-Simulink environment. And the feedback control loop for the NIBB is developed in the digital simulator environment of FPGA: Xilinx Vitis System generator an integrated environment with MATLAB-Simulink. The high-frequency feedback control loop is tested using Hardware-in-the-loop (HIL) FPGA co-simulation emulating the real-time operation of the controller interacting with plant model (WBG converter) designed using the PLECS-blockset.

II. INTEGRATED DESIGN ENVIRONMENT

The design environment for system development is essential to verify the prototype ahead of the experiments by performing different validation tests [3]. It is significant to understand the converter prototype design explicitly as well as minimize the risk of damage and failure.

A. Design and Prototyping Methodology

There are possible methodologies in Fig.2 that can be used for the system prototyping. One of the most used techniques is a fully software-based simulation environment. In this case, the plant model as well as system control are designed and validated in the software environment before experimental prototyping. Another technique involves the hybrid platform consisting of software and hardware system interacting to each other. It has two possibilities: Plant model (power converter) in software and controller model in hardware; the controller model in software and plant model in the hardware.

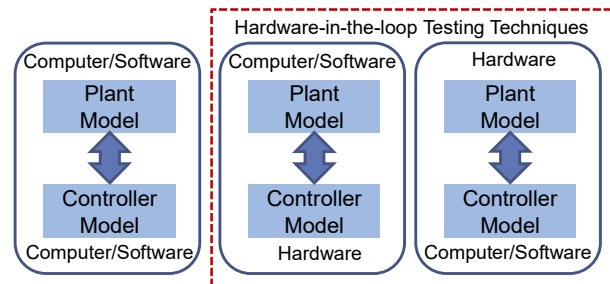


Fig.1 Power electronic system development and prototyping methodology for feedback control system.

The plant model in the hardware means the converter circuit prototype programmed in a hardware.

The plant model (power converter) can be designed and programmed in the hardware and interact with the controller model in the hardware. However, the plant model is still not real hardware that is an experimental prototype and considered as the advanced stage of the plant model (power converter) in software and controller model in hardware.

B. Hybrid Integrated Design Platform

The integrated design platform is defined considering two design environments interfaced with each other. In this case multiple design environment are interfaced within a main platform termed as hybrid integrated design platform. The hybrid integrated platform in Fig. 3a consisting of PLECS specialized for the circuit simulation and digital simulator environment of Vitis Model Composer: system generator by Xilinx for FPGA-based system design. The specialized environment is integrated within MATLAB-Simulink that provides the hybrid platform to interface the controller developed in the system generator to power converter circuit model in PLECS to realize more realistic simulation environment.

The GaN-based model was designed in the PLECS using the thermal model of the devices from the GaN systems. The thermal model of the GaN device GS61004B enhancement mode GaN transistor with a rating of 100V is used to develop a dc-dc converter model using the PLECS blockset. Furthermore, a voltage feedback control is modeled in the

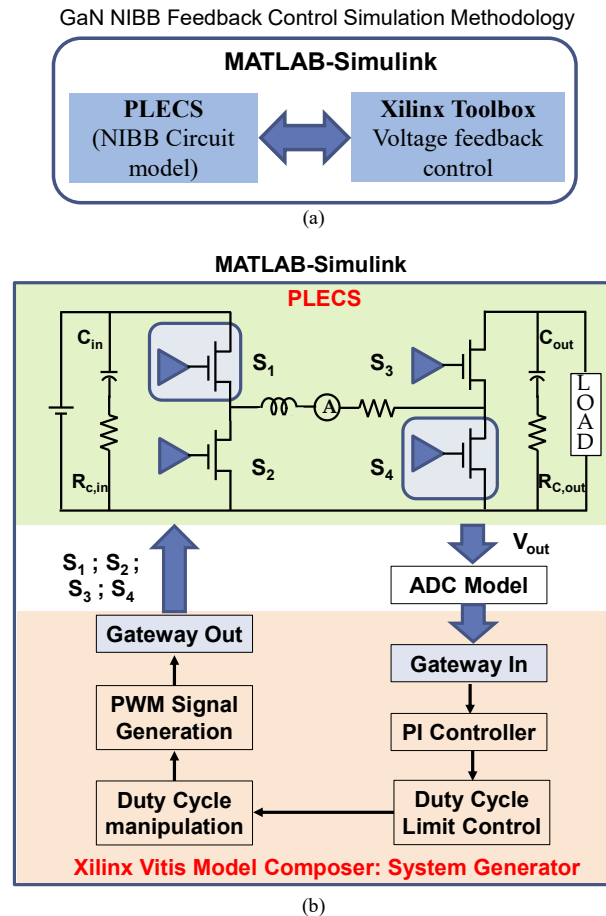


Fig.2 Hybrid integrated design platform.

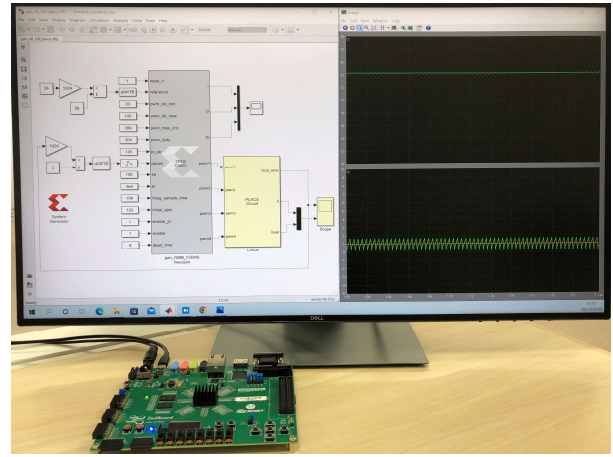


Fig.3 FPGA Hardware-in-the-loop (HIL) co-simulation using Zedboard FPGA Development board and kit.

system generator integrated within the MATLAB-Simulink environment intended to realize high-frequency FPGA-based control of GaN-based converter. The schematic of the controller model in Fig.3b interfaced with the GaN NIBB circuit model in the PLECS in MATLAB-Simulink.

C. Hardware-in-the-loop (HIL) FPGA Co-simulation

The simulation methodology in Fig.1 consists of two fundamental steps: fully software-based simulation and another one is HIL testing. HIL testing is subdivided into two possible undertakings for the controller and the plant model. In this paper controller model in hardware and plant model (power converter) in the software are developed to realize the integrated design platform using the HIL testing.

The high-frequency control model developed in the Xilinx Vitis Model composer system generator interacts with the power converter plant model developed in the PLECS. Further, the developed feedback control model was employed for the HIL testing through FPGA co-simulation. The HIL FPGA co-simulation environment in Fig.3 using the Zeboard FPGA development board and kit that interacts with the power converter model in the software simulation environment.

III. DC-DC CONVERTER SYSTEM

The power converter circuit design involves the power switches and passive components: inductor and capacitor. The rudimentary model of power converter involves the ideal power switches and passive components to design the feedback control development. The non-ideal inductor and capacitor possess the equivalent series resistance (ESR) that affects the performance of the actual converter.

A. Converter Model

The semiconductor power device design requires finite element modeling (FEM) using the technology of computer-aided design (TCAD) [4]. However, this kind of model is not at all pragmatically appropriate for circuit design and system development. Furthermore, the device model required reduced complexity, however, not ideal that would be appropriate enough for the circuit design and control system development. Considering that, the thermal model of the GaN switch for the PLECS is sufficient to achieve that goal.

The passive components comprise the ESR corresponding to the operating frequency. Therefore, it is better to use a resistance in series with the ideal-inductor and ideal-capacitor model in the simulation environment. The dc-dc converter model in PLECS Fig.2b developed with the inductor, input capacitor and output capacitor by incorporating a series resistance.

B. Operating Principle

The GaN-based NIBB circuit topology in Fig.2b consists of four GaN switches as two half-bridge legs, an inductor, and input-output capacitors. In the circuit, the size of the inductor is governed by the current rating, voltage rating and switching period. The switching period for GaN-based NIBB converter is smaller due to high switching-frequency operation, consequently, the inductor size is reduced.

The operating principle of the NIBB converter is categorized based on the working mode: buck and boost mode [5]. Also, the operating mode can be defined considering the switching condition of the power devices as conduction and freewheeling mode. In boost mode, the high-side switch of the input leg is on, and switches of the output side leg are switching. Similarly, the high-side switch of the output leg is on, and switches of the input side leg are switching under the buck mode operation. The switching pattern of the NIBB converter under boost and buck mode is like the boost converter and buck converter respectively.

The pragmatic approach for the NIBB converter is to operate it in buck-boost mode. It is required to achieve a smooth transition between the buck and boost mode by avoiding the dead zone. The devices S2 and S3 are switching under the freewheeling mode and called as the rectification switch. The buck-boost operation is governed by the duty cycle condition of switch S1 and S4 which regulates the output voltage corresponding to the input voltage. This is given by the following equation:

$$v_0 = \frac{d_1}{1 - d_2} v_i ; \quad \begin{cases} d_1 > 1 - d_2 & \text{Boost} \\ d_1 < 1 - d_2 & \text{Buck} \end{cases}$$

d_1 is the duty cycle for the switch S1 and d_2 is duty cycle for the switch S4.

The operating condition of the switches is defined by the duty of the switching sequences correspondingly. The duty of the switching sequences is manipulated through feedback control to regulate the output voltage/current of the converter system. Furthermore, inductor current ripple is a crucial factor to consider for the operating condition of the power converter. The dead-time condition is an additional factor, especially in the case of GaN-based NIBB converter, because of the poor reverse conduction (freewheeling). Dstage 2 and stage 4 involve the reverse conduction of the devices under freewheeling mode that occurred during the dead-time between the higher and lower switches of the corresponding half-bridge legs.

IV. RESULTS AND DISCUSSIONS

NIBB dc-dc converter system designed in PLECS, and voltage feedback control model is developed in FPGA simulator environment: Xilinx Vitis Model composer-system generator. The simulation model was developed considering

the LED application and fixed resistive load is used to replicate the LED load condition. The simulation model was designed for the application with a voltage rating of 24V. The NIBB converter desired to operate with a one-to-one ratio means the 24V output corresponds to the input voltage of 24V. The change in load condition is considered to demonstrate the system's transient behavior.

A. Software Simulation Results (System generator-PLECS)

The simulation results were obtained using the feedback control modeled in the system generator digital simulator of FPGA and power converter circuit in PLECS represented in Fig.2. The simulation is performed for the switching frequency of 500kHz using the fundamental system clock frequency of 200MHz. The sampling time of the MATLAB-Simulink set to 5ns corresponding to the clock frequency of 200MHz.

The simulation was performed considering the transient condition due to changes in the load conditions. The performance of feedback control for GaN NIBB converter in Fig.4a demonstrated for the operation under load condition of 10A Fig.4b; load change from 10A to 5A (at $t=4\text{ms}$) Fig.4c; 5A to 1A (at $t=6\text{ms}$) Fig.4d; 1A to 6A (at $t=8\text{ms}$) Fig.4e.

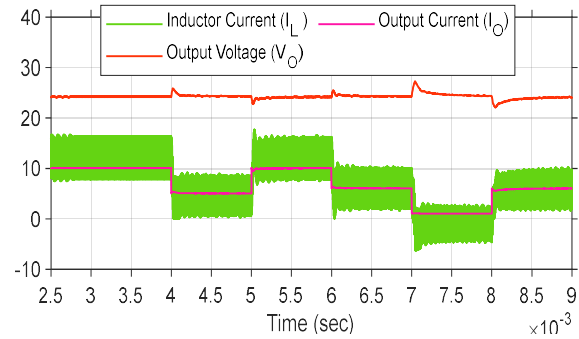


Fig.4a Simulation result of GaN NIBB 24V, load change (10A:1A)

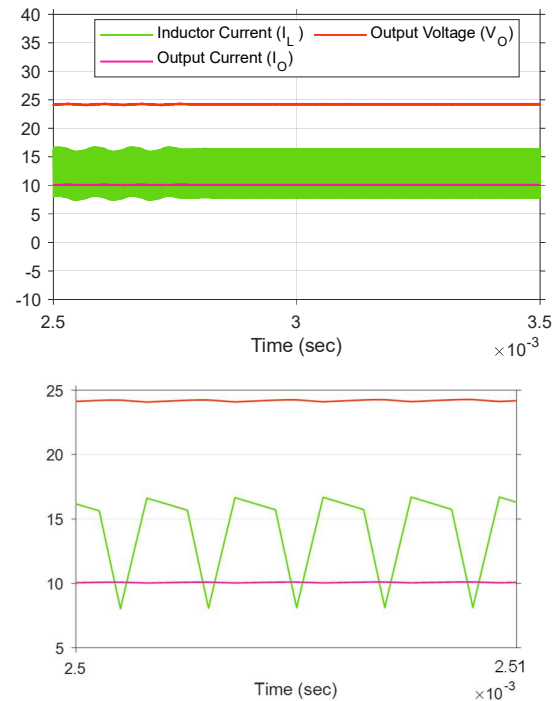


Fig.4b Simulation result of GaN NIBB for output of 24V and 10A.

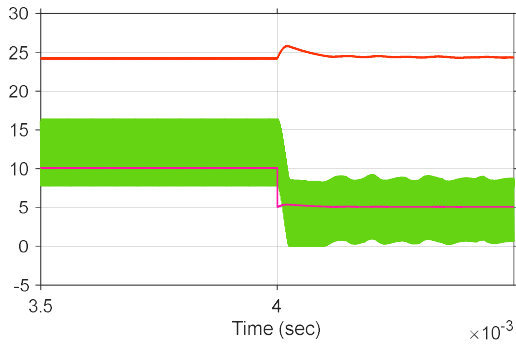


Fig.4c Simulation result of GaN NIBB for ouput of 24V and 5A

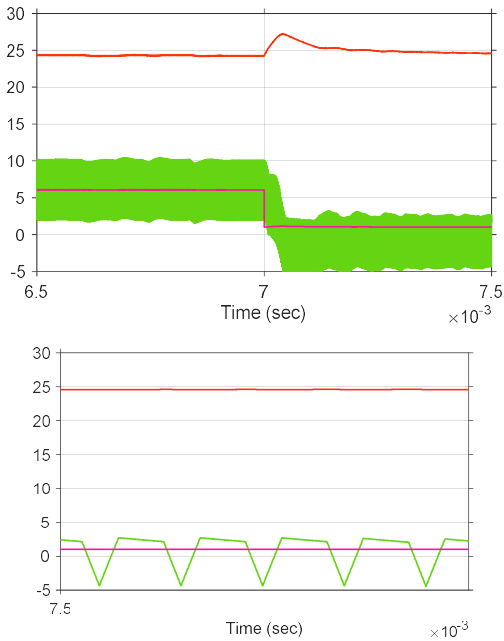


Fig.4d Simulation result of GaN NIBB for ouput of 24V and 1A

B. FPGA HIL Co-simulation Results

The HIL co-simulation was performed using the FPGA Zedboard kit in Fig.3, and results are obtained for the feedback control running inside the FPGA that interacts with the power converter model in PLECS. The co-simulation is running at the system clock frequency of 100MHz, and the power converter is switching at 500kHz. The system clock frequency of 100MHz is sufficient to achieve the PWM resolution for 500kHz switching frequency. The HIL co-simulation is performed considering the similar operation condition of the load as in the case of the software-based simulation. The sampling time of the MATLAB-Simulink

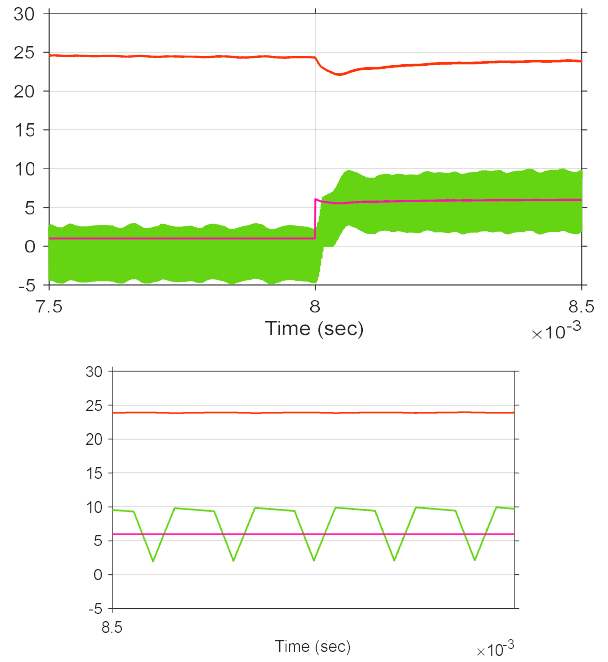


Fig.4e Simulation result of GaN NIBB for ouput of 24V and 6A

and the digital blocks are required to be set to one for the real-time FPGA HIL co-simulation.

The performance of the GaN NIBB converter under the co-simulation; is demonstrated in Fig.5a corresponding to the load condition of 10A Fig.5b; load change from 10A to 5A (at $t=4\text{ms}$) Fig.4c; 1A to 6A (at $t=6\text{ms}$) Fig.4d. The performance of the power converter under HIL co-simulation is close to software-based simulation. However, there are differences during the transient condition and HIL co-simulation exhibit better transient performance with minimized overshoot and undershoot corresponding to changes in the load condition. Furthermore, the shape of inductor current, as well as inductor current ripple, is identical.

C. Experimental Results

The feedback control is authenticated through experimental testing for the GaN-based NIBB power converter. The power converter is operated for the one-to-one output voltage ratio. The dc input voltage of 24V is applied to the converter to achieve the regulated output voltage of

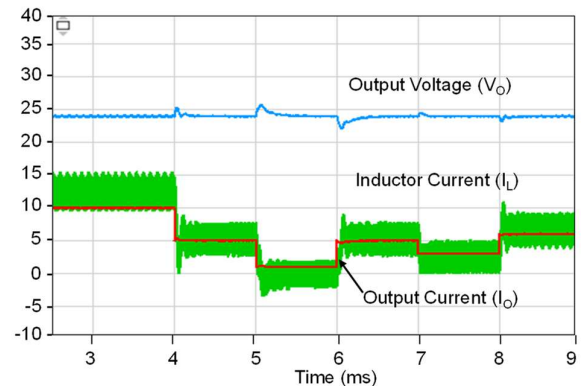


Fig.5a FPGA HIL result of GaN NIBB 24V, load change (10A:1A)

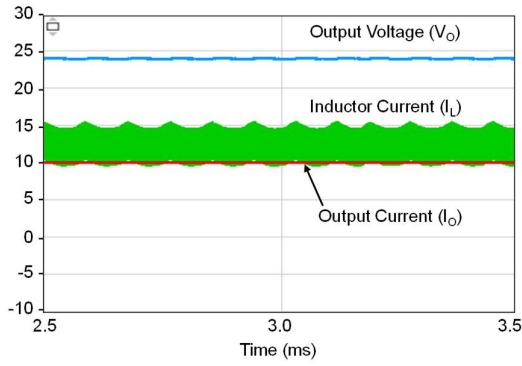


Fig.5b FPGA HIL co-simulation result of GaN NIBB for output of 24V and 10A

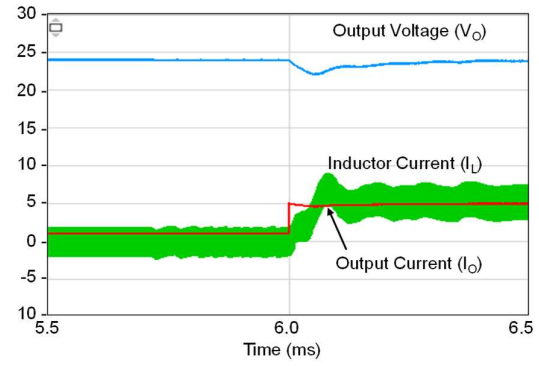
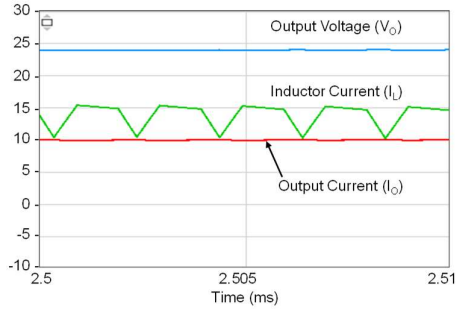
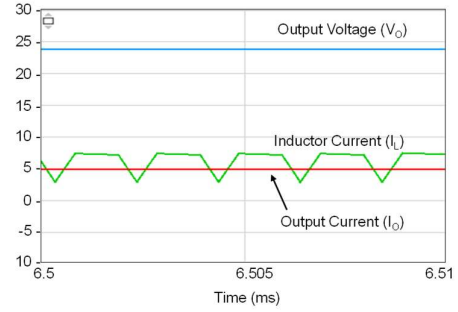


Fig.5d FPGA HIL co-simulation result of GaN NIBB for output of 24V and load change from 1A to 6A.



24V corresponding to operation under different load conditions.

The experimental testing is performed for the validation of the developed feedback control using FPGA with the fundamental frequency of 200MHz, providing the duty resolution of 5ns. The performance of the GaN-based NIBB converter is investigated at 500kHz, deadtime of 50ns considering different load conditions from 1A to 6A. The experimental result is demonstrated in Fig. 6 using the electronic load emulating the LED load condition of 24Ω

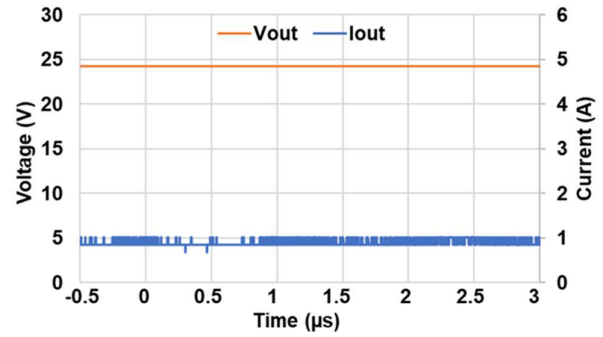


Fig.6 Experimental result of GaN NIBB for output of 24V and 1A

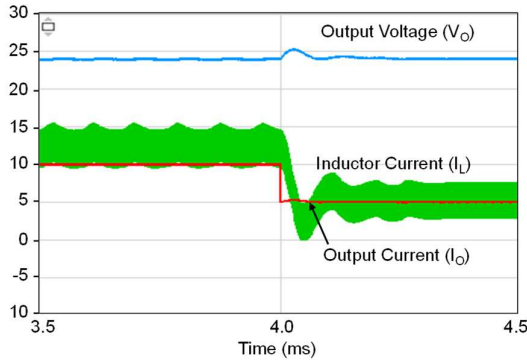
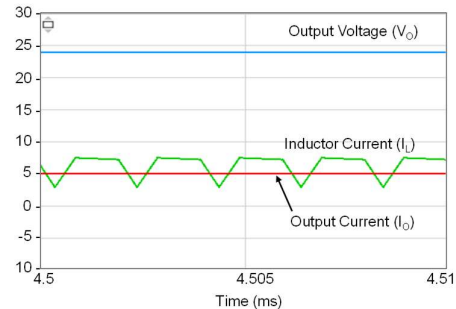


Fig.5c FPGA HIL co-simulation result of GaN NIBB for output of 24V and load change from 10A to 5A.



which results in the output current of approximately 1A with the deadtime condition of 50ns.

The performance of the GaN NIBB power converter is summarized in Table I corresponding to the operation under different load conditions changing from 1A to 6A. The duty cycle corresponding to switch S4 d_o is used as the control parameter to control and regulate the output voltage of the converter. The change from the light load to high load results in the output voltage dip, consequently, d_o is manipulated with the increase in duty cycle through the feedback control to compensate for the dip in the output voltage.

The software-based simulation results for feedback control of the NIBB converter are summarized in Table II considering the operation the same as the experimental test. The performance in terms of voltage regulation corresponding to d_o of switch S4; demonstrates similar trends for the experimental results and software simulation results considering operation under the load condition from 1A to 6A. However, the controller gains are completely different in the case of experimental results. The controller gains for

Table I Experimental test performance of NIBB dc-dc converter for the change in load operating at clock frequency of 200MHz.

Load [Ω]	V _{in} [V]	V _{out} [V]	dφ [ns]
24	24	24.2	355
12	24	24.15	370
8	24	24.1	380
5.9	24	24.1	405
4.7	24	24.1	525
3.9	24	24.1	545

Table II Software-based simulation result of NIBB dc-dc converter for the change in load operating at clock frequency of 200MHz.

Load [Ω]	V _{in} [V]	V _{out} [V]	dφ [ns]
24	24	24.3	400
12	24	24.2	400
8	24	24.25	410
6	24	24.2	450
4.8	24	24.15	540
4	24	24.15	550

Table II FPGA HIL co-simulation result of NIBB dc-dc converter for the change in load operating at clock frequency of 100MHz.

Load [Ω]	V _{in} [V]	V _{out} [V]	dφ [ns]
24	24	23.95	340
12	24	24.15	360
8	24	24.25	470
6	24	24.2	480
4.8	24	24.0	510
4	24	24.15	520

software simulation and FPGA HIL co-simulation the is not the same.

V. CONCLUSION

This paper presented the platform for development and rapid prototyping of the high-frequency controller for wide-band gap power devices. The voltage feedback control design and modeled in the hybrid integrated design platform suitable for circuit simulation and digital controller development using FPGA. The software-based simulation and FPGA HIL co-simulation is demonstrated for the rapid prototyping of the feedback control and authentication before the experimental validation. Finally, the experimental testing of the developed feedback control is presented to validate the converter performance for the developed high-frequency feedback control for operation under different load conditions. The design and prototyping methodology are crucial for FPGA based high-frequency control development considering the WBG devices and straightforward deployment for the industrial applications.

Acknowledgments

Author would like to thank Prof. Alberto Castellazzi for the useful discussion and support for experiments in the laboratory.

VI. REFERENCES

- [1] Y. Xie, P. Brohlin, "Optimizing GaN performance with an integrated driver, Texas Instruments 2016.
- [2] W.Qian, F.Jin, H.Bai, D.Lu, B.Cheng, "Using high-control-bandwidth FPGA and SiC inverters to enhance high-frequency sensorless control in interior permanent magnet synchronous machine", IEEE Access, vol.6, page 42454-42466, 2018.
- [3] Ipsita Mishra, Ravi Nath Tripathi, Vijay Kumar Singh and Tsuyoshi Hanamoto, "A Hardware-in-the-Loop Simulation Approach for Analysis of Permanent Magnet Synchronous Motor Drive", 10th International Conference on Power Electronics (ICPE 2019-ECCE Asia), BEXCO, Busan, Korea, 2019
- [4] R.Kotecha, G.Moreno, B.Mather, S.Narumanchi, "Modeling needs for power semiconductor device and power electronic systems", IEEE IEDM conference, San Francisco USA, 2019.
- [5] X. Weng, X. Xiao, W. He, Y. Zhou, Y. Shen, W. Zhao, Z. Zhao, "Comprehensive comparison and analysis of NIBB and conventional buck boost converters", The journal of Engineering, IET, 2018.